Controllable Generative Recommendation via Guided Token Refinement

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1 PROBLEM AND EVALUATION

Why control at inference time?

A user may shift from relevant results to broader discovery. Items are represented by Semantic IDs whose tokens run from coarse semantics to fine details.

A change in recommendation objective

INITIAL GOAL **Relevance** → AFTER GOAL SHIFT **Diversity**

Autoregressive Semantic ID generation

$c^1 \rightarrow c^2 \rightarrow c^3 \rightarrow c^4$

Tokens are generated coarse to fine. Once an early token is chosen, later predictions follow that fixed prefix; changing it mid-generation may break ID validity.

Limits of other control routes

Training-time Changing the objective may require retraining.

Re-ranking Cannot recover items missing from its candidate pool.

ReGen steers coarse tokens during generation and revisits uncertain details.

1.1 EXPERIMENTAL SETUP

Amazon review datasets (2023)

Dataset	Users	Items
Musical Instruments	57,439	24,587
Industrial & Scientific	50,985	25,848
Video Games	94,762	25,612

Evaluation protocol

Chronological leave-one-out: second-last interaction for validation; last for test.

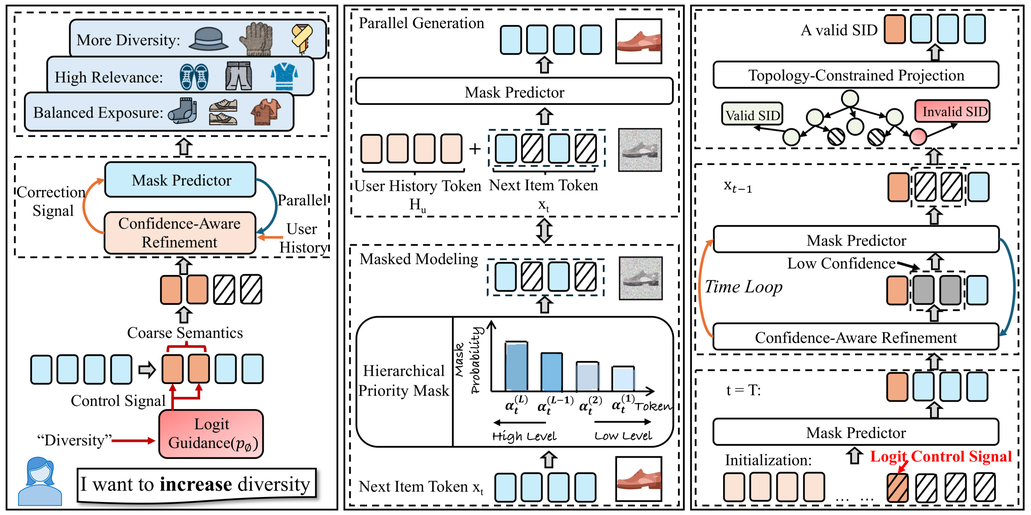
Metrics and comparison methods

Recall (R) and NDCG (N) measure ranking utility. Diversity (D) measures category coverage. All are evaluated at k ∈ {5, 10}.

Compared methods: GRU4Rec, SASRec, BERT4Rec, VQ-Rec, RPG, LLaDA-Rec, ComiRec, Paragon and TIGER-MMR.

Trade-off curves compare methods at similar Diversity@10. Ablations use a fixed diversity-control setting.

2 THE ReGen FRAMEWORK



SID = Semantic ID. Left: control objective. Middle: masked prediction. Right: iterative refinement.
Start from an all-[MASK] ID and user history. Category diversity is evaluated, other controls are conceptual.

2.1 GUIDANCE AND TOKEN REFINEMENT

1 Guide coarse semantics

Downweight coarse tokens common in user history.

$$\tilde{Z}^{(l)} = Z^{(l)} + \lambda s_y^{(l)},\ l \le K$$

Z / Z̃: model / guided scores. y: diversity objective.
s_y: history-frequency penalty. λ: guidance strength.
l: token level. Guide only the first K coarse levels.

2 Repair uncertain tokens

Identify positions that no longer fit the guided context.

$$\rho_t^{(l)} = \max \mathrm{Softmax}(\tilde{Z}_t^{(l)})$$

ρ: highest token probability. t: refinement round.
Re-mask if $\rho < \tau_t$, the round-specific threshold.
Keep guided anchors and reliable tokens.

One partially specified Semantic ID (schematic)

c^1	[MASK]	c^3	[MASK]	→ Mask Predictor
Guided anchor	Re-mask	Reliable	Re-mask	Condition on user history

Repeat prediction, guidance and confidence checks for T rounds.

2.2 VALID CANDIDATES AND LIST SELECTION

Catalog constraints and candidate search

A Trie stores valid catalog ID paths. Before beam expansion, it rules out token choices incompatible with the fixed context. Each round retains up to B = 50 candidate IDs.

From Semantic IDs to a ranked recommendation list

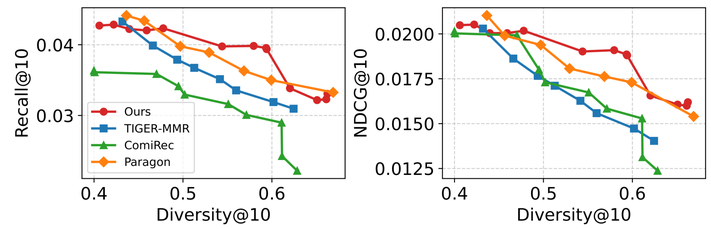
After T rounds, map IDs to items, merge duplicates and filter out items in the user history.
Pick k items greedily by the highest controlled score + μ × the number of new categories added.
Each item uses its best generated-ID score. μ sets the weight of category coverage.

Training masks coarse tokens less often, so the model learns to reconstruct finer details from them.
At inference, change the control without updating the pretrained backbone.

3 RESULTS

Accuracy–diversity trade-off · Ours = ReGen

Musical Instruments



At D@10 ≈ 0.54, ReGen R@10 / N@10: 0.0398 / 0.0190.
TIGER-MMR: 0.0351 / 0.0163.

Industrial & Scientific

At D@10 ≈ 0.56, ReGen R@10 / N@10: 0.0351 / 0.0173.
TIGER-MMR: 0.0235 / 0.0109.

3.1 COMPONENT ABLATION

Instruments	R@10	D@10	Val@10
ReGen (full)	**0.0339**	**0.5606**	**99.97%**
w/o Guidance	0.0355	0.4851	99.97%
w/o Refinement	0.0130	0.7321	99.97%
w/o Topology	0.0136	0.0001	27.63%

Table III: fixed diversity-control setup. One component changes per row.
Val@10: percentage of generated IDs that map to real items.

3.2 CONCLUSION

ReGen offers a favorable accuracy–diversity trade-off under moderate guidance.
Guidance steers coarse semantics; refinement helps preserve relevance.

Experiments evaluate category diversity.
Other control objectives need separate evaluation.
Effectiveness depends on the quality of the Semantic ID codebook.

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DM410